INCH-POUND

MIL-A-82836(OS)
5 February 1990
SUPERSEDING
WS 26953
14 December 1987

MILITARY SPECIFICATION

ADHESIVE, EPOXY RESIN, STRUCTURAL BONDING, NON-ASBESTOS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of non-asbestos structural bonding adhesive consisting of a filled epoxy resin and an amine curing agent.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-81533 1,1,1-Trichloroethane (Methyl Chloroform) Inhibited, Vapor Degreasing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1106, Chicago, IL 60606-5945.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 214 Sieve Analysis of Granular Metal Powders (DoD adopted)

ASTM D 1652 Epoxy Content of Epoxy Resins (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the adhesive shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material.** The material shall be a two-part adhesive consisting of a gray, aluminum-powder-filled, thixotropic epoxy resin in paste form (part A), and an amber, amine-type curing agent in liquid form (part B).

3.2.1 **Asbestos.** The adhesive material shall contain no asbestos (see 6.2).

3.2.2 **Aluminum powder particle size.** The particle size of the aluminum powder used as filler material shall be such that a minimum of 93 percent of the powder shall pass through a number 325 mesh sieve.

3.3 **Chemical and physical properties.** The chemical and physical properties shall conform to table I.

3.4 **Formulation.** The adhesive formulation shall be as follows:

a. Part A: 100 ± 1 parts by weight

b. Part B: 33 ± 1 parts by weight

3.5 **Age of material.** The material shall be stored at a maximum temperature of 0°F and shall be shipped within 12 months of manufacture (see 6.2).

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TABLE I. Chemical and physical properties.

Property	Ingredients				Mixed Material	
	Part A		Part B			
	Min	Max	Min	Max	Min	Max
Filler content, percent by weight	43	49	---	---	---	---
Epoxide content, grams per equivalent	170	210	---	---	---	---
Titratable nitrogen, percent by weight	---	---	11.8	14.1	---	---
Tensile adhesion, steel-to-steel, pounds per square inch	---	---	---	---	2600	---
Working life, minutes	---	---	---	---	40	---

3.6 **Safety.**

3.6.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.6.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.7 **Shelf life.** The adhesive shall have a shelf life of 12 months after date of shipment when stored at 40°F maximum. The maximum cumulative time out of controlled storage shall be 30 days.

3.7.1 **Shelf life extension.** The shelf life expiration date of an individual lot may be extended for an additional 12 months provided the adhesive, upon retest, conforms to the tensile adhesion requirement specified in table I.

3.8 **Workmanship.** The adhesive component materials shall be free from agglomeration, foreign matter, separation, dilution, and contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except

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as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in table II.

4.4.1 Lot. A lot of part A and part B shall consist of the material of each part compounded and mixed in a single container in one mixing period without change in process or materials and submitted for inspection at one time.

4.4.2 Sampling.

4.4.2.1 For visual examination. Sampling for visual examination, to determine conformance to section 5, shall consist of a random selection of at least 10 percent of the lot. Sampling for visual examination, to determine conformance to 3.2 and 3.8, shall consist of the sample selected in accordance with 4.4.2.2.

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TABLE II. Quality conformance inspection.

Tests and Inspections	Requirement	Test Method
Visual examination	3.2, 3.2.1, 3.5, 3.8 and Section 5	4.5.1
Filler content (part A)	3.3	4.5.2
Epoxide content (part A)	3.3	4.5.3
Titratable nitrogen (part B)	3.3	4.5.4
Tensile adhesion strength (mixed material)	3.3	4.5.5
Working life (mixed material)	3.3	4.5.6
Aluminum particle size	3.2.2	4.5.7

4.4.2.2 **For tests.** Sampling for tests shall consist of a minimum of 1 quart of part A with sufficient quantity of part B for proper composition, with each part selected at random from the lot.

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Visual examination of the samples selected in 4.4.2.1 shall be conducted to determine conformance to section 5. Visual examination of the sample selected in 4.4.2.2 shall be conducted to determine conformance to 3.2 and 3.8. Contractor certification shall be examined for conformance to 3.2.1 and 3.5 (see 6.2).

4.5.2 **Filler content (part A).** Filler content of part A shall be determined in accordance with the following:

a. Weigh 0.7 to 0.8 gram (g) of sample, to the nearest 0.1 milligram (mg), into a beaker, and add 10 to 20 milliliters (ml) of chlorobenzene. Stir until all lumps are dissolved and any suspended insoluble matter is finally dispersed.

b. Filter the sample solution through a tared and matted medium-porosity, sintered-glass crucible into a vacuum flask, using

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vacuum. Wash residue with 50 ± 10 ml chlorobenzene. Retain the filtrate for the epoxide contents determination.

c. Dry the crucible and contents for at least 30 minutes at 300 ± 20°F. Place in a desiccator, cool to room temperature, and weigh to the nearest 0.1 mg.

d. Calculation:

$$\text{Filler content (percent)} = \frac{(W1 - F)}{W} \times 100$$

Where: W1 = weight of crucible and residue, g

W = weight of sample, g

F = weight of crucible and matting, g

e. Record the average of at least two determinations.

4.5.3 Epoxide content (part A). Epoxide content of part A shall be determined in accordance with ASTM D 1652. If the filler content has been determined, the filtrate may be used for the sample.

4.5.4 Titratable nitrogen (part B). Titratable nitrogen of part B shall be determined in accordance with 4.5.4.1 through 4.5.4.4 (or alternatively as specified in 4.5.4.5).

4.5.4.1 Equipment.

a. Calomel reference electrode.

b. Glass electrode, all-purpose.

c. pH meter.

4.5.4.2 Reagents.

a. Acetic acid, glacial, American Chemical Society (ACS) reagent grade.

b. Acetic anhydride, ACS reagent grade.

c. Potassium acid phthalate, primary standard.

d. Perchloric acid, ACS reagent grade.

4.5.4.3 Perchloric acid solution. Perchloric acid solution, 0.1 Normal (N) shall be prepared as follows:

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a. Place 250 ± 50 ml of glacial acetic acid in a 1000-ml volumetric flask.

b. Add 8 to 9 ml of 70-percent perchloric acid or 10 to 11 ml of 60-percent perchloric acid, and mix.

c. Add 20 ml of acetic anhydride, and mix.

d. Dilute to volume with glacial acetic acid, and mix.

e. Stopper the flask, and let stand for at least 24 hours.

f. Weigh 0.45 to 0.50 g of dried potassium acid phthalate, to the nearest 0.1 mg, into a 250-ml beaker.

g. Add 50 to 100 ml of glacial acetic acid. Stir and heat gently until all potassium acid phthalate has dissolved.

h. Cool to room temperature, immerse electrodes in the sample (solution), and titrate potentiometrically with perchloric acid.

i. Perform a blank determination.

j. Calculation:

$$\text{Normality of perchloric acid} = \frac{W \times (4.897)}{V1 - V2}$$

Where: W = weight of potassium acid phthalate, g

V1 = volume of perchloric acid required to titrate sample, ml

V2 = volume of perchloric acid required to titrate blank, ml

4.897 = reciprocal of milliequivalent weight of potassium acid phthalate

k. Normality between duplicate standardizations must not differ by more than 0.001.

4.5.4.4 **Procedure.** Titratable nitrogen content shall be determined as follows:

a. Weigh a 0.2- to 0.3-g sample, to the nearest 0.1 mg, into a 250-ml beaker.

b. Add 50 to 100 ml of glacial acetic acid. If necessary, warm the solution to completely dissolve the sample.

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c. If warming was used to dissolve the sample, allow the solution to cool to room temperature.

d. Place electrodes in the solution, and titrate potentiometrically with standardized perchloric acid solution.

e. Perform a blank determination.

f. Calculation:

$$\text{Titratable nitrogen (percent)} = \frac{(V1 - V2)\ (N)\ (0.014)}{W} \times 100$$

Where: V1 = volume of perchloric acid solution required to titrate sample, ml

V2 = volume of perchloric acid solution required to titrate blank, ml

N = normality of perchloric acid solution

W = weight of sample, g

0.014 = milliequivalent weight of nitrogen

g. Record the average of at least two determinations.

4.5.4.5 Alternative titration procedure. Alternatively, the standardization and the sample titration may be performed visually using crystal violet indicator, providing that the sample coloration does not interfere with the observation of the blue-green endpoint. In case of dispute, the potentiometric titration procedure shall govern.

4.5.5 Tensile adhesion test (mixed material). The tensile adhesion test shall be performed in accordance with the following:

4.5.5.1 Cleaning and preparation of adhesion discs.

a. Place steel adhesion discs (see figure 1) in a vapor degreaser, and spray with methyl chloroform conforming to MIL-T-81533. Leave the discs in the degreaser 5 to 10 minutes.

b. Remove the discs from the degreaser and check for stains. If stains appear, repeat step 4.5.5.1 a.

c. Grit blast the bonding surface of each disc with 100 to 200 mesh zirconium silicate or equivalent.

d. Blow grit from bond surface using clear, dry nitrogen.

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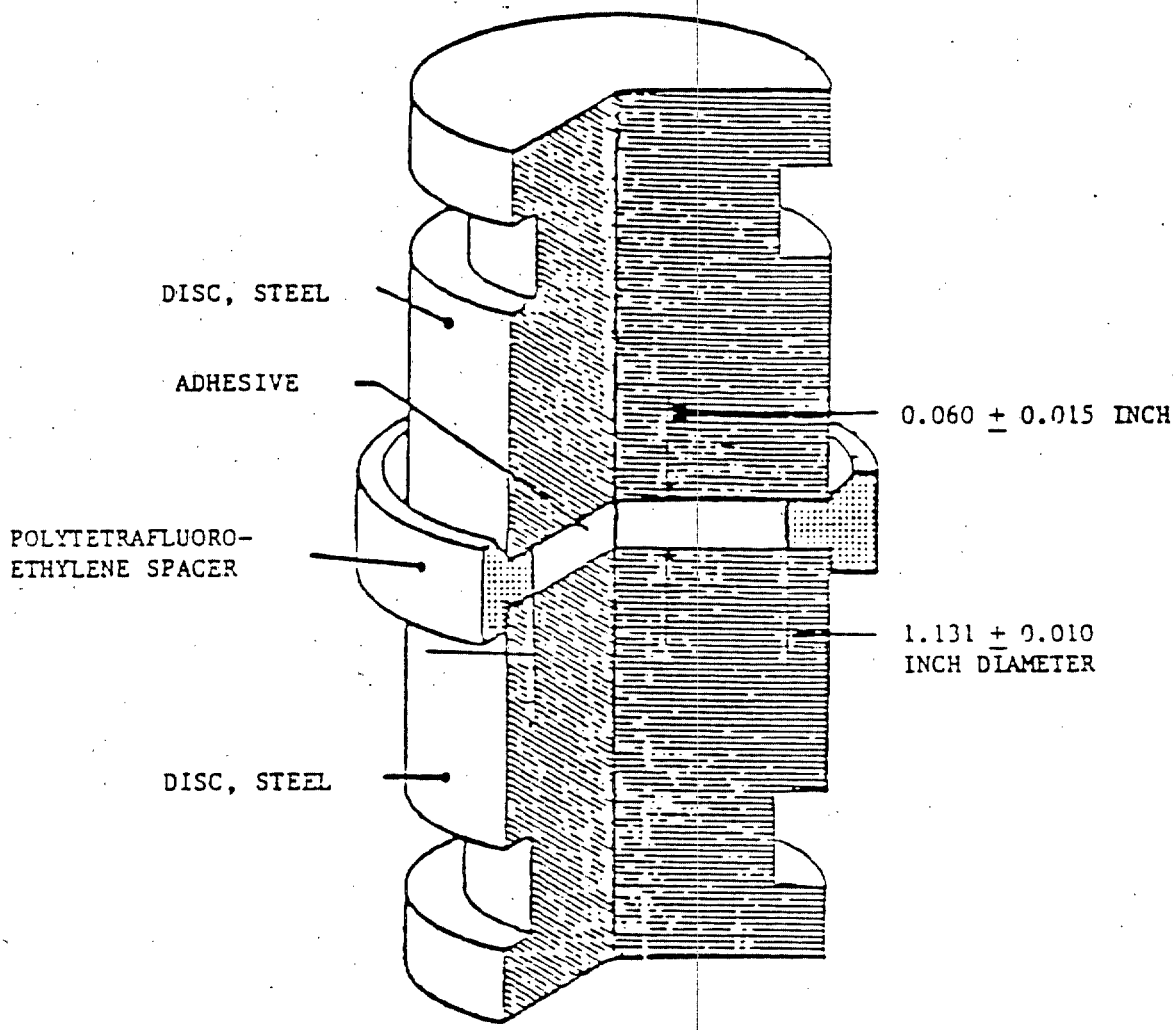


FIGURE 1. Tensile adhesion test specimen assembly.

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e. Repeat steps a, b, c, and d if more than 6 hours elapses before bonding operation or if the bond surface has been touched with fingers.

4.5.5.2 Preparation of test specimens.

a. Place an adhesion disc spacer on the bonding surface of one adhesion disc.

b. Apply sufficient mixed adhesive to completely cover the glue line ring. Ensure that the entire bonding surface is wetted.

c. If air bubbles have formed, puncture with a sharp, clean instrument.

d. Wet the bonding surface of a second adhesion disc with adhesive.

e. Place the bonding surface of the second adhesion disc on the material in the spacer with a rotary motion, and press down until the top disc is seated firmly on the spacer and all excess material has been forced out of the spacer cavity.

f. Remove any excess material which could affect test results.

g. Wrap the assembled tensile adhesion specimens from end-to-end with shrink tape to ensure the adhesion discs are properly seated and aligned during cure. Using cellophane tape, attach one end of the shrink tape to the disc surface and the other end to itself. After several wraps of the shrink tape, apply hot air from a heat gun to shrink the tape. (See figure 1 for typical tensile adhesion test specimen assembly.)

h. Identify the test specimen as required for test traceability.

i. Cure the test specimen 2.0 to 2.5 hours at 180 $\pm$ 10°F.

j. Remove the tape.

4.5.5.3 Test procedure. A total of five specimens shall be pulled to failure in accordance with the following:

a. After conditioning each test specimen at 75 $\pm$ 5°F for at least 30 minutes, place each specimen in an Instron Tensile Tester Model TTC-M1, or equivalent, equipped with Type F load cell, or equivalent.

b. Apply an indicated crosshead speed of 0.05 inch per minute in tension at ambient temperature until failure of specimen.

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c. Pull a total of five specimens to failure. Determine the maximum force applied to each specimen and record this directly as pounds per square inch (psi).

NOTE: Since the bonded area of the specimen is 1.0 square inch, record this reading directly as psi.

d. Any values indicating rejection shall be reexamined. If a test specimen shows more than 10 percent voids, as determined visually, that value may be disregarded.

e. Record the reason for rejecting any test specimen.

f. Accept the raw material lot if a minimum of three test specimens comply with the tensile adhesion requirement of table I.

g. Record the adhesive strength of each specimen and the average to the nearest psi. Record the percent adhesive or cohesive failure.

4.5.6 **Working life (mixed material).** Working life of the mixed material shall be determined in accordance with the following:

a. Condition part A and part B at 77 ± 5°F for at least 10 hours.

b. Thoroughly mix adhesive in a ratio of 100 ± 1 parts by weight (pbw) of part A to 33 ± 1 pbw of part B and in sufficient quantity to provide a 400- to 500-g sample weight. Stir vigorously with a spatula until uniform consistency is obtained.

c. Start timing at completion of mixing.

d. Allow the sample to stand undisturbed in a water bath maintained at 77 ± 5°F until tested for working life.

e. Test for spreadability with a spatula at 40 minutes after start of timing. If spreadable to a uniform thickness of 0.05 ± 0.01 inch, record the adhesive as "conforming." If not spreadable to a uniform thickness of 0.05 ± 0.01 inch, record the adhesive as "nonconforming."

f. Record results of a minimum of one determination.

4.5.7 **Aluminum powder particle size.** Particle size of the aluminum powder filler shall be determined by sieve analysis in accordance with ASTM B 214. Only a number 325 mesh sieve shall be used.

4.6 **Inspection of packaging.** The adhesive containers and container markings shall be visually examined to verify compliance with section 5.

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5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The adhesive shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.7. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Shelf life expiration date.

e. Storage conditions.

f. Toxicity warning.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The adhesive is intended for general use in structural bonding of rocket motor components.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

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c. Whether first article inspection is required (see 3.1 and 6.3)

d. DOD FAR 52.246-15 (see 4.5.1).

e. Material Safety Data Sheets Required (see 3.6.2 and 6.4)

f. Special marking, if other than as specified in 5.2.2

6.3 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 8040-N153)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82836	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
ADHESIVE, EPOXY RESIN, STRUCTURAL BONDING, NON-ASBESTOS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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